Ramrao Adik Institute of Technology, Nerul, Navi Mumbai

Question Bank – 5 (Module-5)

Academic year: 2025-2026

Subject: Embedded System Class: S.E. Sem : III

1. Draw detailed diagram of Seven Segment Display (SSD) interfacing with 8051 microcontroller and write suitable program for the same.
2. Discuss in detail the Embedded case study of "Digital Camera".
3. Discuss in detail the Embedded case study of "Washing Machine".
4. Draw detailed diagram of DC Motor interfacing with 8051 microcontroller and write suitable program for the same.
5. Draw detailed diagram of Stepper Motor interfacing with 8051 microcontroller and write suitable program for the same.
6. Draw detailed diagram of Liquid Crystal Display (LCD) interfacing with 8051 microcontroller and write suitable program for the same.
7. Draw detailed diagram of Hex Keyboard interfacing with 8051 microcontroller and write suitable program for the same.
8. Explain what is "Wave Drive, Full Step and Half Step" sequence with reference to stepper motor interfacing.
9. Assume that switch SW is connected to the pin P2.0 and P1.0 is connected to one of the input of DC Motor and P1.1 is connected to another input of DC Motor. Write a program to monitor the status of SW, if SW = 1, rotate the dc motor in a clockwise direction, or if SW = 0, reverse the direction of the motor.
10. Explain the difference between Common Cathode and Common Anode SSDs.